

Browser Acceptance Specification

Functional interaction metrics and physical-WebGL production authorization

mdstats

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Contents

1	Purpose and status	1
2	Public contracts	1
3	Functional policy	1
4	Production authorization	2
5	Input payload	2
6	Determinism and safety	2
7	Focused validation	2

1 Purpose and status

Module: `mdstats.plotting.density_browser_acceptance`

Status: normative and implemented; production authorization requires evidence from a physical WebGL renderer.

This module evaluates browser-runner evidence after a valid HTML artifact has been generated. It does not alter geometry or budgets.

2 Public contracts

```
classify_webgl_renderer(vendor, renderer)
BrowserAcceptancePolicy
BrowserAcceptanceReport
evaluate_browser_acceptance(validation, policy=None)
```

Renderer classes are:

```
physical
software
unavailable
```

3 Functional policy

The default policy evaluates:

- first complete frame time;
- scripted camera-orbit rate;
- representative trace-toggle latency;

- WebGL context loss;
- JavaScript heap use when available.

A functional pass and production-default authorization are separate. A managed or software renderer may provide useful functional evidence without authorizing the profile as a production default.

4 Production authorization

When `require_physical_webgl_for_production=True`, the validation payload must expose a physical renderer and pass every functional threshold. Missing vendor/renderer information yields `unavailable`, not an inferred physical pass.

5 Input payload

The evaluator consumes the JSON-compatible output of the browser-validation runner. Malformed `browser` or `metrics` sections raise `GraphAdapterError`.

6 Determinism and safety

Acceptance evaluation is pure and does not modify:

- geometry;
- trace order;
- legend groups;
- density fields;
- budgets;
- HTML artifacts.

7 Focused validation

Tests must cover:

- renderer classification;
- threshold boundaries;
- context-loss behavior;
- missing heap and renderer data;
- functional pass with production failure;
- full production authorization;
- report serialization.